IV & LATE extensions

Econ5106

LATE

Counterfactual distributions

Multiple instruments

Variable treatment intensity

Covariates

Conclusion

IV with heterogeneous effects

When estimating the effect of d_i on y_i with IV z_i the standard textbook case presents the outcome equation with homogenous effects

$$y_i = \alpha + \beta d_i + u_i$$

But we can link observed outcome y_i to potential outcomes $y_i(d)$ as follows

$$\begin{aligned} y_i &= \underbrace{E[y_i(0)]}_{\alpha} + \underbrace{(y_i(1) - y_i(0))}_{\beta_i} d_i + \underbrace{y_i(0) - E[y_i(0)]}_{u_i} \\ &\equiv \alpha + \beta_i d_i + u_i \end{aligned}$$

What does IV identify when treatment effects are heterogenous?

Heterogeneous potential outcome set-up

In IV framework we can think of the instrument as initiating a causal chain, where z_i affects the variable of interest d_i which in turn affects y_i

Keeping this in mind we can adopt the potential outcome set-up:

- $d_i(z)$ is treatment status at instrument value $z_i = z$
- $y_i(d, z)$ is outcome of individual i when receiving treatment $d_i = d$ and instrument value $z_i = z$

We can now define various causal effects:

- $y_i(1, z_i) - y_i(0, z_i)$
- $y_i(d_i, 1) - y_i(d_i, 0)$
- $d_i(1) - d_i(0)$
- $y_i(d_i(1), 1) - y_i(d_i(0), 0)$

To identify causal effects, we rely on four assumptions.

Heterogeneous potential outcome set-up

The first assumption in the heterogeneous effects set-up is random assignment (instrument is independent to the **potential** outcome and **potential** treatment)

Random assignment

$y_i(d, z), d_i(z) \perp z_i \, \forall \, d, z$

This is sufficient to identify average causal effect of z_i on y_i (and of z_i on d_i):

$$
\begin{aligned}
E[y_i|z_i = 1] - E[y_i|z_i = 0] \\
&= E[y_i(d_i(1), 1)|z_i = 1] - E[y_i(d_i(0), 0)|z_i = 0] \\
&= E[y_i(d_i(1), 1) - y_i(d_i(0), 0)]
\end{aligned}
$$

Heterogeneous potential outcome set-up

The second assumption in the heterogeneous effects set-up is the exclusion restriction:

Exclusion restriction

$y_i(d, 1) = y_i(d, 0)$

This states that any effect of z_i on y_i must be via an effect of z_i on d_i

The **exclusion restriction** is often expressed by omitting z_i in equation of interest: $y_i = \alpha + \beta \cdot d_i + u_i$

Assumptions 1 +2 are sometimes implicitly stated together as instrument exogeneity: $u_i \perp z_i$

Random assignment + exclusion restriction $\implies$ *instrument exogeneity*

Heterogeneous potential outcome set-up

The third assumption in the heterogeneous effects set up is relevance: the existence of a first stage.

First stage

$E[d_i(1) - d_i(0)] \neq 0$

Which requires the instrument z_i to have some effect on the average probability of treatment

Heterogeneous potential outcome set-up

The fourth assumption in the heterogeneous effects set-up is monotonicity

Monotonicity

$d_i(1) \geq d_i(0) \quad \forall i \; (or\, vice\, versa)$

Which says that all those affected by the instrument are affected in the same direction

Local Average Treatment Effect (LATE)

Under the following assumptions:

1. Random assignment: $y_i(d, z), d_i(z) \perp z_i \; \forall d, z$
   - gives the causal effect of z on d (1st stage) and y (reduced form)
2. Exclusion Restriction: $y_i(d, 1) = y_i(d, 0)$
   - so that the causal effect of z on y is only due to the effect of z on d
3. Monotonicity: $d_i(1) \geq d_i(0) \; \forall i$, or vice versa
   - to avoid offsetting effects
4. Relevance: $E[d_i(1) - d_i(0)] \neq 0$
   - because we need treatment variation in the sample

The Wald estimand using instrumental variable z_i recovers the *Local Average Treatment Effect*:

$$\beta_{IV} = E[\beta_i | d_i(1) = 1, d_i(0) = 0]$$

the average treatment effect for compliers: those with treatment changed by the instrument

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the average treatment effect for compliers: those with treatment changed by the instrument

Local Average Treatment Effect (LATE)

The LATE is the averaged effect of the treatment on outcomes for individuals who would be treated if $z_i = 1$, but who would not be treated if $z_i = 0$

To see why this is so, we can divide the population into four groups:

1. **Compliers**: $d_i(1) = 1$ and $d_i(0) = 0$;
2. **Always-takers**: $d_i(1) = 1$ and $d_i(0) = 1$;
3. **Never-takers**: $d_i(1) = 0$ and $d_i(0) = 0$;
4. **Defiers**: $d_i(1) = 0$ and $d_i(0) = 1$;

Local Average Treatment Effect (LATE)

This can be represented using the following table.

	$d_i(1) = 1$	$d_i(1) = 0$
$d_i(0) = 0$	**Compliers**	**Never-takers**
$d_i(0) = 1$	**Always-takers**	**Defiers**

Local Average Treatment Effect (LATE)

Can observations of D and Z tell us which groups they belong to?

	$D = 0$	$D = 1$
$Z = 0$	**Compliers, Never-takers**	**Defiers, Always-takers**
$Z = 1$	**Defiers, Never-takers**	**Compliers, Always-takers**

Not directly! More on this in a minute.

Local Average Treatment Effect (LATE)

Proof that β_{IV} is LATE

We saw that (by independence)

$$E[y_i|z_i = 1] - E[y_i|z_i = 0] = E[y_i(d_i(1), 1) - y_i(d_i(0), 0)]$$

The average causal effect of z_i on y_i can be written as weighted average of the causal effects of the four sub-populations:

$$\begin{aligned} E[y_i(d_i(1),1) - y_i(d_i(0),0)] = \\ E[y_i(d_i(1), 1) - y_i(d_i(0), 0)|\text{Complier}] \times P(d_i(1) = 1, d_i(0) = 0) \\ +E[y_i(d_i(1), 1) - y_i(d_i(0), 0)|\text{Never taker}] \times P(d_i(1) = 0, d_i(0) = 0) \\ +E[y_i(d_i(1), 1) - y_i(d_i(0), 0)|\text{Always taker}] \times P(d_i(1) = 1, d_i(0) = 1) \\ +E[y_i(d_i(1), 1) - y_i(d_i(0), 0)|\text{Defier}] \times P(d_i(1) = 0, d_i(0) = 1) \end{aligned}$$

Local Average Treatment Effect (LATE)

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The average causal effect of z_i on y_i can be written as weighted average of the causal effects of the four sub-populations:

$$\begin{aligned}
&E[y_i(d_i(1), 1) - y_i(d_i(0), 0)] = \\
&E[y_i(d_i(1), 1) - y_i(d_i(0), 0)|\text{Complier}] \times P(d_i(1) = 1, d_i(0) = 0) \\
&+E[\underbrace{y_i(d_i(1), 1) - y_i(d_i(0), 0)}_{=y(0,1)-y(0,0)=0}|\text{Never taker}] \times P(d_i(1) = 0, d_i(0) = 0) \\
&+E[\underbrace{y_i(d_i(1), 1) - y_i(d_i(0), 0)}_{=y(1,1)-y(1,0)=0}|\text{Always taker}] \times P(d_i(1) = 1, d_i(0) = 1) \\
&+E[y_i(d_i(1), 1) - y_i(d_i(0), 0)|\text{Defier}] \times \underbrace{P(d_i(1) = 0, d_i(0) = 1)}_{=0}
\end{aligned}$$

Local Average Treatment Effect (LATE)

Proof that β_{IV} is LATE

By monotonicity $d_i(1) \geq d_i(0)$, which implies that there are no defiers.

$$E[y_i|z_i = 1] - E[y_i|z_i = 0]$$
$$= E[y_i(1) - y_i(0)|\text{Complier}] \times P(d_i(1) = 1, d_i(0) = 0)$$

and by independence and monotonicity we can show that

$$E[d_i|z_i = 1] - E[d_i|z_i = 0] = E[d_i(1) - d_i(0)] = P(d_i(1) = 1, d_i(0) = 0)$$

From this it follows that the Wald estimand is equal to the average treatment effect on the compliers

$$\frac{E[y_i|z_i = 1] - E[y_i|z_i = 0]}{E[d_i|z_i = 1] - E[d_i|z_i = 0]}$$
$$= \frac{E[y_i(1) - y_i(0)|\text{Complier}] \times P(d_i(1) = 1, d_i(0) = 0)}{P(d_i(1) = 1, d_i(0) = 0)}$$
$$= E[y_i(1) - y_i(0)|\text{Complier}]$$

where relevance guarantees the denominator of the middle term is non-zero.

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$$\frac{E[y_i|z_i = 1] - E[y_i|z_i = 0]}{E[d_i|z_i = 1] - E[d_i|z_i = 0]}$$
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where relevance guarantees the denominator of the middle term is non-zero.

LATE

With heterogeneous effects ...

IV estimates the average causal effect for compliers.

Different valid instruments for same causal relation therefore estimate different target parameters (different groups of compliers)

- Overidentifying restrictions test (Sargan test) might reject even if all instruments are valid.

Is the LATE useful?

Once we allow for heterogeneous effects, all we can estimate with IV is the effect of treatment among compliers.

- This is an unknown subset of the data.
  - Treated units are a mix of always takers and compliers.
  - Control units are a mix of never takers and compliers.
  - You can never know if a *particular individual* is a complier!
- Without further assumptions, $LATE \neq ATT \neq ATE$.
- Complier group depends on the instrument:
  - with heterogeneity different IVs will lead to different estimands.
- How much we care largely depends on our question and how it relates to the instrument.
  - LATE can be interesting. So can be ATT. So can be ATE. So can be *none of these*.
  - “Policy-relevant treatment effects" (Heckman and Vytlacil, 2001)
- The traditional framework assumes that the effect is constant

Who are the compliers?

We cannot identify individual compliers because we can never observe both $d_i(0)$ and $d_i(1)$

- those with $z_i = 1$ and $d_i = 1$ can be compliers or always-takers
- those with $z_i = 0$ and $d_i = 0$ can be compliers or never-takers

	$D = 0$	$D = 1$
$Z = 0$	**Compliers, Never-takers**	**Always-takers**
$Z = 1$	**Never-takers**	**Compliers, Always-takers**

Who are the compliers?

But we can identify characteristics of the compliers as a group!

The size of the complier group is the Wald 1st-stage:

$$\Pr(d_i(1) > d_i(0)) = E[d_i|z_i = 1] - E[d_i|z_i = 0]$$

Or among the treated (Bayes' rule):

$$\begin{aligned}\Pr(d_i(1) > d_i(0)|d_i = 1) &= \frac{\Pr(d_i = 1|d_i(1) > d_i(0))\Pr(d_i(1) > d_i(0))}{\Pr(d_i = 1)} \\ &= (E[d_i|z_i = 1] - E[d_i|z_i = 0])\frac{\Pr(z_i = 1)}{\Pr(d_i = 1)}\end{aligned}$$

We also describe their covariates (Bayes' rule again):

$$\begin{aligned}P(X_i = x|d_i(1) > d_i(0)) &= \frac{P(d_i(1) > d_i(0)|X_i = x)}{P(d_i(1) > d_i(0))}P(X_i = x) \\ &= \frac{E[d_i|z_i = 1, X_i = x] - E[d_i|z_i = 0, X_i = x]}{E[d_i|z_i = 1] - E[d_i|z_i = 0]}P(X_i = x)\end{aligned}$$

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\end{aligned}$$

Randomized trials with one-sided noncompliance

When non-compliance is one-sided, then the LATE is equal to the ATT.

Think of a randomized experiment:

- Z_i = randomized treatment assignment
- D_i = actual treatment taken

One-sided noncompliance: only those assigned to treatment (control) can actually take the treatment (control). Or

$$Pr[D_i = 1 | Z_i = 0] = 0$$

Maybe this is because only those treated actually get pills or only they are invited to the job training location, etc.

Benefits of one-sided noncompliance

One-sided noncompliance no “always-takers" and since there are no defiers,

- Treated units must be compliers.
- ATT is the same as the LATE.

$$
\begin{aligned}
&E[Y_i|Z_i=1]-E[Y_i|Z_i=0]\\
=&E[Y_i(0)+(Y_i(1)-Y_i(0))D_i|Z_i=1]-E[Y_i(0)|Z_i=0]\\
=&E[Y_i(0)|Z_i=1]+E[(Y_i(1)-Y_i(0))D_i|Z_i=1]-E[Y_i(0)|Z_i=0]\\
=&E[Y_i(0)]+E[(Y_i(1)-Y_i(0))D_i|Z_i=1]-E[Y_i(0)]\\
=&E[(Y_i(1)-Y_i(0))|D_i=1,Z_i=1]Pr(D_i=1|Z_i=1)\\
=&E[(Y_i(1)-Y_i(0))|D_i=1]Pr(D_i=1|Z_i=1)
\end{aligned}
$$

Wald Estimator: one-sided noncompliance

Wald Estimator

$$
\begin{aligned}
LATE &= \frac{E[Y_i|Z_i=1]-E[Y_i|Z_i=0]}{E[D_i|Z_i=1]-E[D_i|Z_i=0]} \\
&= \frac{E[(Y_i(1)-Y_i(0))|D_i=1]Pr(D_i=1|Z_i=1)}{Pr(D_i=1|Z_i=1)-Pr(D_i=1|Z_i=0)} \\
&= E[(Y_i(1)-Y_i(0))|D_i=1] = ATT
\end{aligned}
$$

LATE extensions

Until now we considered the IV model with heterogeneity in the simple case of

- average effects
- one instrument
- binary treatment
- no covariates

What happens when we relax these assumptions?

LATE

Counterfactual distributions

Multiple instruments

Variable treatment intensity

Covariates

Conclusion

Counterfactual distributions

Imbens & Rubin (1997) show that we can estimate more than average causal effects

They show how to recover the complete marginal distributions of the outcome

- under different treatments for the compliers
- under the treatment for the always-takers
- without the treatment for the never-takers

These results may be helpful to think about external validity

Counterfactual distributions

First introduce some notation

$$C_i = n \iff D_i^1 = D_i^0 = 0$$
$$C_i = a \iff D_i^1 = D_i^0 = 1$$
$$C_i = c \iff D_i^1 = 1, D_i^0 = 0$$
$$C_i = d \iff D_i^1 = 0, D_i^0 = 1$$

For the different combinations of Z and D, we know the following:

		D	
		0	1
Z	0	n, c	a
	1	n	a, c

Counterfactual distributions

Distribution of types

Since Z is random assigned, we know that the distribution of types a, n, c is **the same** for each value of Z and in the population as a whole

Therefore, this...

		D	
		0	1
Z	0	n, c	a
	1	n	a, c

...implies the following:

$$p_a = \Pr(D = 1 | Z = 0)$$
$$p_n = \Pr(D = 0 | Z = 1)$$
$$p_c = 1 - p_a - p_n$$

Counterfactual distributions

Identifying distributions

Let's use the following notation for the observed marginal distribution of y given z and d:

$$f_{zd}(y) \equiv f(y|Z = z, D = d)$$

Let g_n and g_a, denote marginal distributions of y for n, a

and let $c(d)$ denote the marginal for compliers with treatment d.

Therefore, this...

		D	
		0	1
Z	0	n, c	a
	1	n	a, c

...implies the following:

$$\begin{aligned}
f_{10}(y) &= g_n(y) \\
f_{01}(y) &= g_a(y) \\
f_{00}(y) &= g_{c(0)}(y) \cdot (p_c/(p_c + p_n)) \\
&\quad + g_n(y) \cdot (p_n/(p_c + p_n)) \\
f_{11}(y) &= g_{c(1)}(y) \cdot (p_c/(p_c + p_a)) \\
&\quad + g_a(y) \cdot (p_a/(p_c + p_a))
\end{aligned}$$

Counterfactual distributions

Example

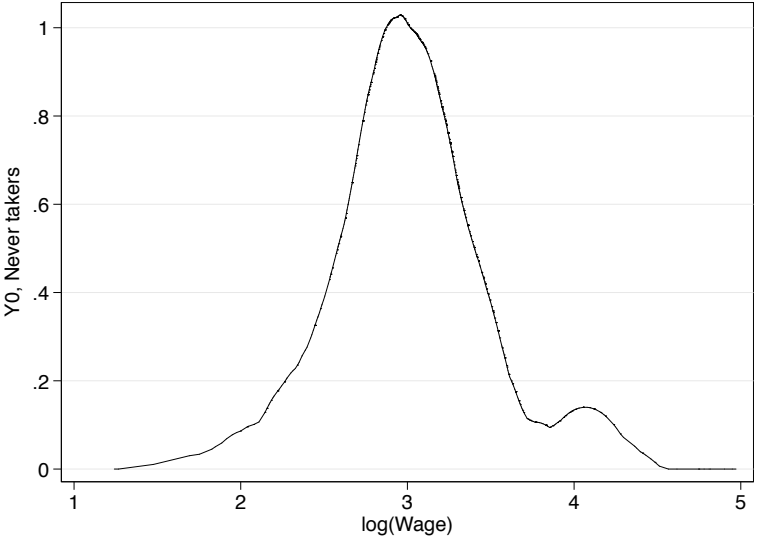
To illustrate the above, consider Dutch data

- Lottery outcome as instrument of medical school completion
  - $D = 1$ if completed medical school
  - $Z = 1$ if offered medical school after successful lottery

```
. ta z d
           |            d
         z |         0          1 |     Total
-----------+----------------------+----------
         0 |       269        187 |       456
         1 |        71        949 |     1,020
-----------+----------------------+----------
     Total |       340      1,136 |     1,476
```

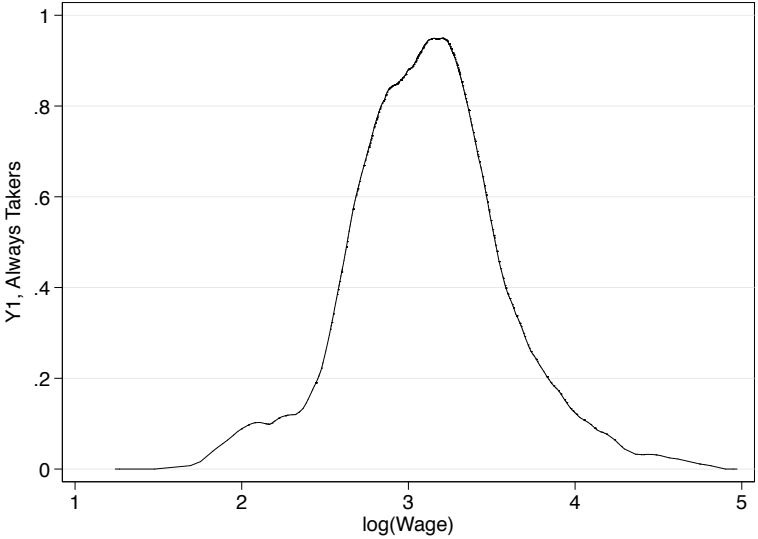
Counterfactual distributions

$f_{10}(y) = g_n(y)$



Counterfactual distributions

$f_{01}(y) = g_a(y)$



Counterfactual distributions

We have seen that we can estimate p_a, p_n, p_c and also $g_n(y)$ (=$f_{10}(y)$) and $g_a(y)$ (=$f_{01}(y)$)

By rearranging the following

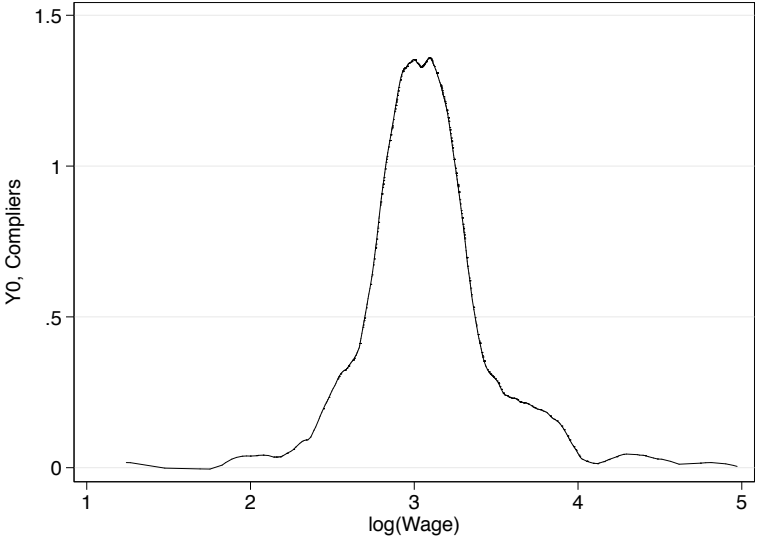
$$\begin{aligned} f_{00}(y) &= g_{c0}(y)\cdot(p_c/(p_c+p_n))+g_n(y)\cdot(p_n/(p_c+p_n)) \\ f_{11}(y) &= g_{c1}(y)\cdot(p_c/(p_c+p_a))+g_a(y)\cdot(p_a/(p_c+p_a)) \end{aligned}$$

we can back out the counterfactual distributions for the compliers:

$$\begin{aligned} g_{c0}(y) &= f_{00}(y)\cdot(p_c+p_n)/p_c - f_{10}(y)\cdot p_n/p_c \\ g_{c1}(y) &= f_{11}(y)\cdot(p_c+p_a)/p_c - f_{01}(y)\cdot p_a/p_c \end{aligned}$$

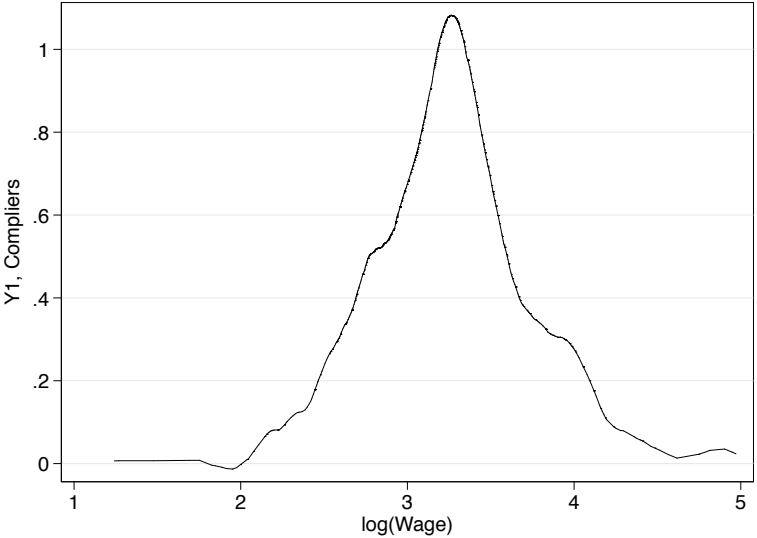
Counterfactual distributions

$g_{c0}(y) = f_{00}(y) \cdot (p_c + p_n)/p_c - f_{10}(y) \cdot p_n/p_c$

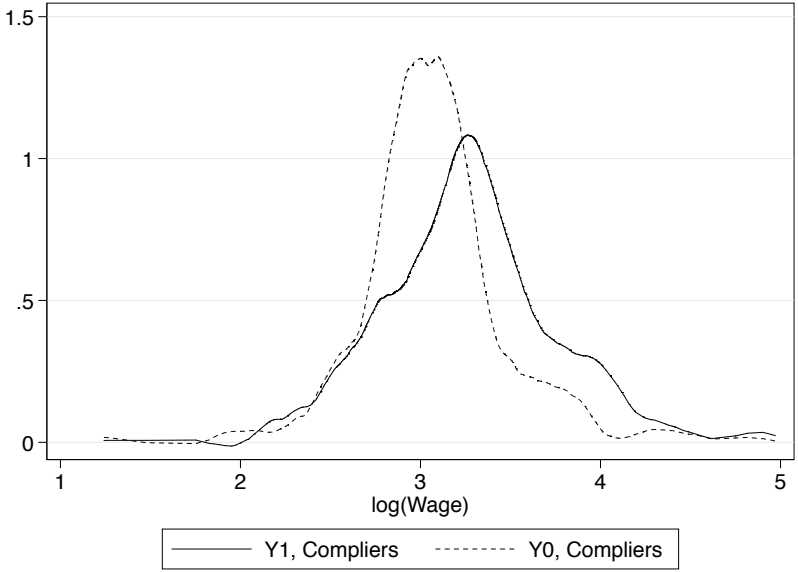


Counterfactual distributions

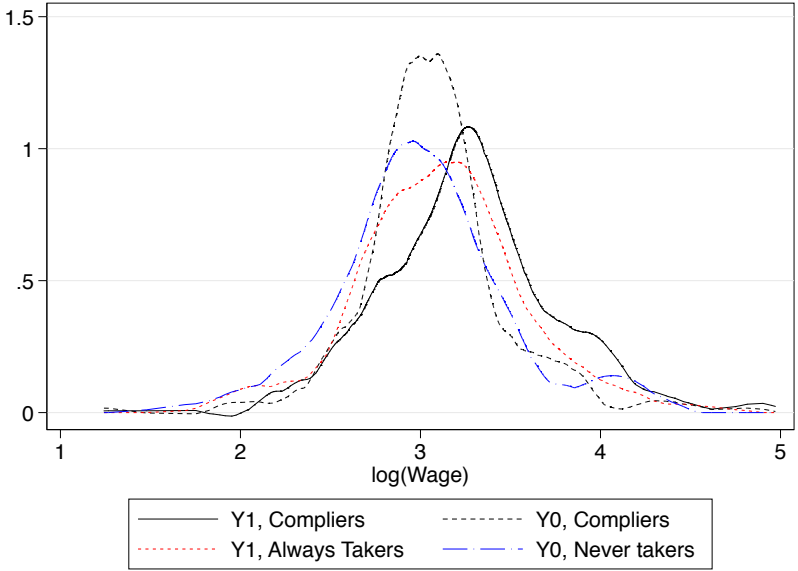
$g_{c1}(y) = f_{11}(y) \cdot (p_c + p_a)/p_c - f_{01}(y) \cdot p_a/p_c$



Counterfactual distributions



Counterfactual distributions



Counterfactual distributions

We can also show that

$$E[Y^1|C=c] = \frac{E[Y \cdot D|Z=1] - E[Y \cdot D|Z=0]}{E[D|Z=1] - E[D|Z=0]}$$

and

$$E[Y^0|C=c] = \frac{E[Y \cdot (1-D)|Z=1] - E[Y \cdot (1-D)|Z=0]}{E[1-D|Z=1] - E[1-D|Z=0]}$$

LATE

Counterfactual distributions

Multiple instruments

Variable treatment intensity

Covariates

Conclusion

LATE with multiple instruments

Assume we have 2 mutually exclusive binary instruments

We can then estimate two different LATEs:

$$\beta_{Z_j} = \frac{cov(Y, Z_j)}{cov(D, Z_j)}$$
$$= E[Y^1 - Y^0 | D^{Z_j=1} - D^{Z_j=0} = 1]$$

In practice you may need to combine the instruments to obtain a more efficient estimator using 2SLS.

The 2SLS estimator is

$$\beta_{2SLS} = \frac{cov(Y, \hat{D})}{cov(D, \hat{D})}$$

where $\hat{D} = \pi_1 Z_1 + \pi_2 Z_2$; π_1 and π_2 are "marginal" first stage effects.

You should be asking: what is β_{2SLS}?

LATE with multiple instruments

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LATE with multiple instruments

Expanding $\hat{D}$ in the expression for β_{2SLS} gives

$$\begin{aligned}\beta_{2SLS} &= \pi_1 \frac{cov(Y, Z_1)}{cov(D, \hat{D})} + \pi_2 \frac{cov(Y, Z_2)}{cov(D, \hat{D})} \\ &= \pi_1 \frac{cov(D, Z_1)}{cov(D, \hat{D})} \frac{cov(Y, Z_1)}{cov(D, Z_1)} + \pi_2 \frac{cov(D, Z_2)}{cov(D, \hat{D})} \frac{cov(Y, Z_2)}{cov(D, Z_2)} \\ &= \psi \beta_{Z_1} + (1 - \psi) \beta_{Z_2}\end{aligned}$$

where

$$\psi \equiv \frac{\pi_1 cov(D, Z_1)}{\pi_1 cov(D, Z_1) + \pi_2 cov(D, Z_2)}$$

is the relative strength of Z_1 in the first stage

The 2SLS estimate is an instrument-strength weighted average of the instrument specific LATEs

LATE

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Variable treatment intensity

Assume treatment is no longer binary but varies in its level

$$S_i \in \{0, 1, 2, \ldots, J\}$$

such as for example years of schooling.

We can then define potential outcomes indexed by the *level* of treatment

$$Y_i^S$$

Potential treatments (schooling level) are as before indexed by the value of the instrument

$$S_i^Z$$

so that with a binary instrument the observed level of schooling is

$$S_i = Z_i S_i^1 + (1 - Z_i) S_i^0$$

Variable treatment intensity

The observed outcome

$$Y_i = \sum_{s=0}^{J} Y_i^s 1[S_i = s] = Y_i^0 + \sum_{s=1}^{J} (Y_i^s - Y_i^{s-1}) I[S_i \geq s]$$

The average effect of the s-th year of schooling is

$$E[Y^s - Y^{s-1}]$$

and we have now J different treatment effects

In a linear model these are assumed to be the same: $E[Y^s - Y^{s-1}] = \beta$

$$Y = \alpha + \beta S + u$$

Variable treatment intensity

As before we need to make an independence assumption

$$Y^{s,z}, S^z \perp Z \quad \forall s, z$$

and an exclusion restriction

$$Y^{s,z} = Y^s$$

We further need a monotonicity assumption

$$S^1 \geq S^0$$

and instrument relevance

$$E[S^1 - S^0] \neq 0$$

Variable treatment intensity

Example with 3 levels

With three treatment intensities $S \in \{0, 1, 2\}$, monotonicity implies

$$I[S_1 \geq l] - I[S_0 \geq l] \in \{0, 1\} \text{ for } l \in \{0, 1, 2\}$$

since Z must increase each individual's treatment level. So CDFs reveal how individual treatment levels are sensitive to Z:

$$\begin{aligned}
\Pr(S_1 \geq l) - \Pr(S_0 \geq l) &= E(I[S_1 \geq l]) - E[I(S_0 \geq l)] \\
&= \Pr(I[S_1 \geq l] > I[S_0 \geq l]) \\
&= \Pr(S_1 \geq l > S_0)
\end{aligned}$$

if this probability is greater than 0, then the instrument affects the incidence of treatment level l.

$$\begin{aligned}
&E[S|Z = 1] - E[S|Z = 0] \\
&\quad = [\Pr(S_1 \geq 1|Z = 1) - \Pr(S_0 \geq 1|Z = 0)] \\
&\quad + [\Pr(S_1 \geq 2|Z = 1) - \Pr(S_0 \geq 2|Z = 0)] \\
&\quad = \Pr(S_1 \geq 1 > S_0) + \Pr(S_1 \geq 2 > S_0)
\end{aligned}$$

Variable treatment intensity

Example with 3 levels

With three treatment intensities $S \in \{0, 1, 2\}$ we observe

$$Y = Y^0 + (Y^1 - Y^0)I[S \geq 1] + (Y^2 - Y^1)I[S \geq 2]$$

Using this we can expand the reduced form as follows

$$\begin{aligned} E[Y|Z=1] - E[Y|Z=0] &= E[(Y^1 - Y^0)(I[S_1 \geq 1] - I[S_0 \geq 1])] \\ &\quad + E[(Y^2 - Y^1)(I[S_1 \geq 2] - I[S_0 \geq 2])] \\ &= E[Y^1 - Y^0 | S_1 \geq 1 > S_0] \Pr(S_1 \geq 1 > S_0) \\ &\quad + E[Y^2 - Y^1 | S_1 \geq 2 > S_0] \Pr(S_1 \geq 2 > S_0) \end{aligned}$$

Variable treatment intensity

Average Causal Response

We can now define

$$\omega_s = \frac{\Pr(S_1 \geq s > S_0)}{\sum_{j=1}^{J} \Pr(S_1 \geq j > S_0)}$$

and express the Wald estimate as follows

$$\frac{E[Y|Z=1] - E[Y|Z=0]}{E[S|Z=1] - E[S|Z=0]} = \sum_{s=1}^{J} \omega_s E[Y^s - Y^{s-1}|S_1 \geq s > S_0]$$

which Angrist and Imbens call the average causal response (ACR).

Variable treatment intensity

Average Causal Response

We cannot estimate $E[Y^s - Y^{s-1} | S_1 \geq s > S_0]$ for the different local complier groups

What we can do is estimate their weights in the ACR, since

$$\begin{aligned}
\Pr(S_1 \geq l > S_0) &= \Pr(S_1 \geq l) - \Pr(S_0 \geq l) \\
&= \Pr(S_0 < l) - \Pr(S_1 < l) \\
&= \Pr(S < l | Z = 0) - \Pr(S_1 < l | Z = 1)
\end{aligned}$$

which allows us to estimate ω_s

Variable treatment intensity

Example

Angrist & Krueger (1991) use quarter of birth as an instrument for schooling

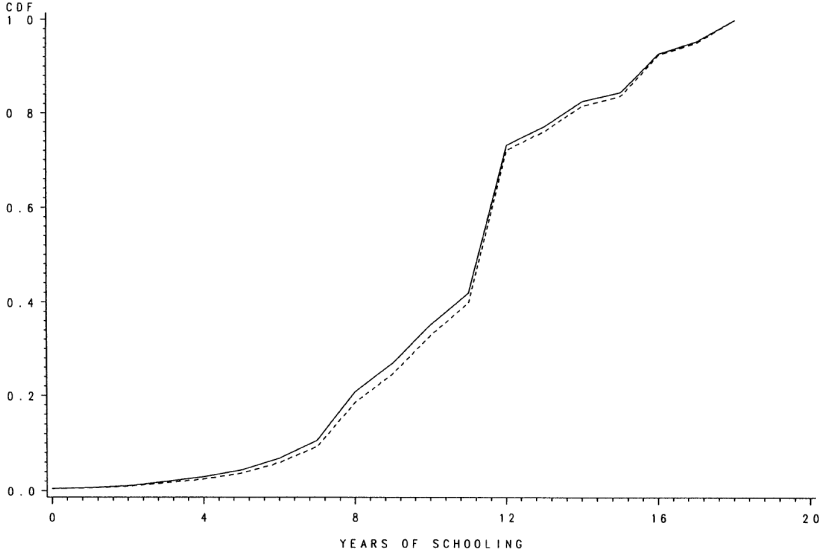
- S years of schooling
- $Z = 1$ if born in the 4th quarter, $Z = 0$ if born in the 1st quarter

How does the Wald estimator weighs the average unit causal response

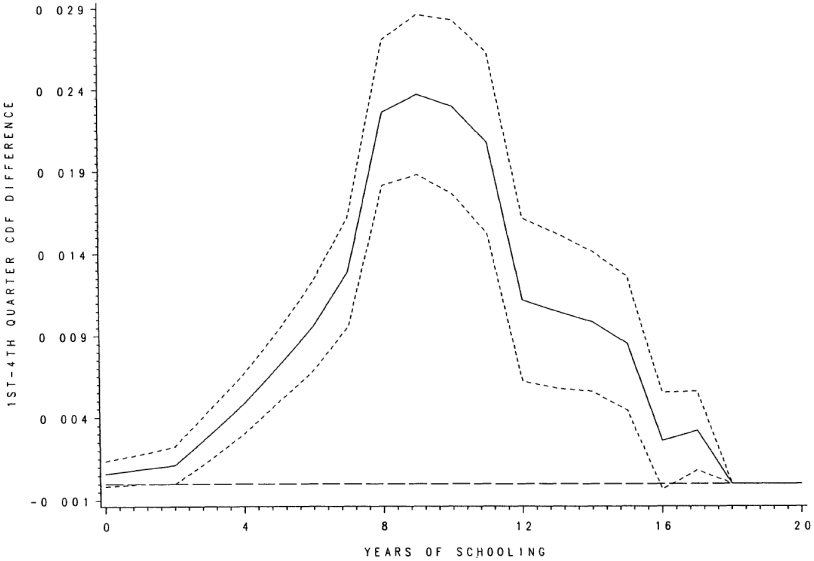
$$E[Y^s - Y^{s-1} | S_1 \geq s > S_0]$$

for the complier at the different points s?

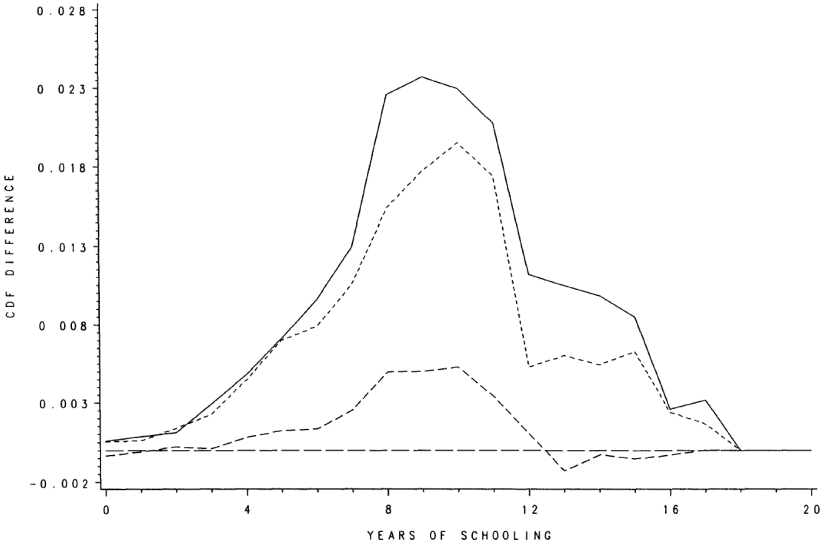
Example, Schooling CDF by QoB (= 1, 4)



Differences in Schooling CDF by QoB (= 1, 4)



Differences in Schooling CDF: 4vs1, 4vs2, 4vs3



LATE

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IV with Covariates

You may also want to include covariates:

- if Z is randomly assigned to increase precision, or
- because we need them to motivate exogeneity:

$$Y^{z,d}, D^z \perp Z|X$$

Suppose we can estimate stratified LATEs

$$\begin{aligned}\beta(x) &= \frac{E[Y|Z=1, X=x] - E[Y|Z=0, X=x]}{E[D|Z=1, X=x] - E[D|Z=0, X=x]} \\ &= E[Y^1 - Y^0|D^1 - D^0 = 1, X=x]\end{aligned}$$

We want to go from here to some population averaged LATE

Which one would we like to have?

IV with Covariates

What does a saturated 2SLS estimation gives us?

$$\begin{aligned} Y_i &= \beta D_i + \alpha_x + e_i \\ D_i &= \pi_x Z_i + \gamma_x + u_i \end{aligned}$$

i.e. x-dummies in both stages, and x-specific first-stage coefficients

Angrist & Imbens (1995) show that

$$\beta = E[\beta(x)\omega(x)]$$

where $\beta(x)$ is the x-specific LATE, and

$$\omega(x) = \frac{\sigma^2_{\hat{D}}(x)}{E[\sigma^2_{\hat{D}}(x)]} = \frac{\pi_x^2 \sigma_Z^2(x)}{E[\pi_x^2 \sigma_Z^2(x)]}$$

The weighting thus depends on the square of the local (to x) complier share and instrument variance

Note: recovering a true LATE will require some reweighting!

IV with Covariates

Abadie κ-weighting

What if full saturation is infeasible? Abadie (2003) showed that we can "restrict" a (semi-)parametric regression to model complier outcomes, by reweighting!

$$E[\kappa_0 g(Y,X)] = E[g(Y^0,X)|D^1 > D^0]\Pr(D^1 > D^0)$$
$$E[\kappa_1 g(Y,X)] = E[g(Y^1,X)|D^1 > D^0]\Pr(D^1 > D^0)$$
$$E[\kappa g(Y,D,X)] = E[g(Y,D,X)|D^1 > D^0]\Pr(D^1 > D^0)$$

where

$$\kappa_0 = (1-D)\frac{(1-Z)-\Pr(Z=0|X)}{\Pr(Z=0|X)\Pr(Z=1|X)}$$
$$\kappa_1 = D\frac{Z-\Pr(Z=1|X)}{\Pr(Z=0|X)\Pr(Z=1|X)}$$
$$\kappa = \kappa_0\Pr(Z=0|X) + \kappa_1\Pr(Z=1|X)$$
$$= 1 - \frac{D(1-Z)}{\Pr(Z=0|X)} - \frac{(1-D)Z}{\Pr(Z=1|X)}$$

IV with Covariates

Abadie κ-weighting

Consider the three terms in κ

$$\kappa = 1 - \frac{D(1-Z)}{\Pr(Z=0|X)} - \frac{(1-D)Z}{\Pr(Z=1|X)}$$

By monotonicity, those with $D(1-Z)=1$ are always-takers because they have $d(0)=1$.

Those with $(1-D)Z=1$ are never-takers since $d(1)=0$.

κ removes means for always-takers and never-takers from marginal means.

What is left is average for compliers.

IV with Covariates

Abadie κ-weighting

Consequently we can estimate a causal CEF with weighted least squares

$$(\alpha, \beta) = \arg\min_{a,b} E[\kappa_i (y_i - g(a \cdot d_i + x_i' b))^2]$$

for any function $g(\cdot)$

Think of this as running a regression in the complier population

To implement the above we need an estimate of $\Pr(Z = 1|X)$ (and $\Pr(Z = 0|X)$)

If $\Pr(Z = 1|X)$ can be consistently estimated with a LPM and $g(x_i' b) = a \cdot d_i + x_i' b$, then Abadie's approach gives the same estimate for α as 2SLS (Proposition 5.1, Abadie 2003)

- 2SLS probably approximates $E[Y|D, X, D^1 > D^0]$

(Abadie provides asymptotic results, but s.e.'s can be bootstrapped)

LATE

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I understand/can apply...

How IV addresses selection bias

IV with

- homogenous treatment effects
- heterogeneous treatment effects

Local Average Treatment Effect (LATE)

- always takers, never takers, compliers, defiers
- monotonicity assumption
- independence assumption
- exclusion restriction
- instrument relevance (1st stage)

Interpret LATE

I understand/can apply...

Identify compliers

Estimate not only effects but also counterfactual outcome distributions

Average causal response

- IV produces weighted averages of ordered treatment effects

Covariates are naturally incorporated and 2SLS probably approximates causal CEF for compliers

- but weighting not always innocuous

Sketch of proof Abadie (2003) results

Implicitly condition everywhere on X, then

$$\begin{aligned} E[Y|c]P(c) &= E[Y] - E[Y|a]P(a) - E[Y|n]P(n) \\ &= E[Y] - E[Y|D=1, Z=0]P(D=1|Z=0) \\ &\quad -E[Y|D=0, Z=1]P(D=0|Z=1) \end{aligned}$$

where the second equality follows from

$$\begin{aligned} E[Y|a]P(a) &= E[Y|D=1, Z=0]P(D=1|Z=0) \\ E[Y|n]P(n) &= E[Y|D=0, Z=1]P(D=0|Z=1) \end{aligned}$$

Sketch of proof Abadie (2003) results

since

$$
\begin{aligned}
E[D(1-Z)Y] &= E[Y|D=1,Z=0]P(D=1|Z=0)P(Z=0) \\
E[Z(1-D)Y] &= E[Y|D=0,Z=1]P(D=0|Z=1)P(Z=1)
\end{aligned}
$$

we get

$$
\begin{aligned}
E[Y|c]P(c) &= E[Y]-E[D(1-Z)Y]/P(Z=0)-E[Z(1-D)Y]/P(Z=1) \\
&= E[Y\cdot(1-\frac{D(1-Z)}{P(Z=0)}-\frac{Z(1-D)}{P(Z=1)})]=E[\kappa Y]
\end{aligned}
$$

Sketch of proof Abadie (2003) results

we can now integrate over X and use Bayes' theorem to get the result

$$
\begin{aligned}
E[Y|c] &= \int E[Y|X,c]f(X|c)dX \\
&= \int \frac{1}{P(c|X)} E[\kappa Y|X]f(X|c)dX \\
&= \int \frac{1}{P(c,X)/f(X)} E[\kappa Y|X]\frac{P(X,c)}{P(c)}dX \\
&= \int \frac{1}{P(c)} E[\kappa Y|X]f(X)dX \\
&= \frac{1}{P(c)} \int E[\kappa Y|X]f(X)dX \\
&= \frac{1}{P(c)} E[\kappa Y]
\end{aligned}
$$